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(An Autonomous Institute)



DEPARTMENT OF ELECTRONICS & TELECOMMUNICATION

ENGINEERING

PROJECT

REPORT ON

“Surveillance Robot with Real-Time Video Streaming using ESP32-CAM”

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PROJECT REPORT

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Abstract:

The “Surveillance Robot with Real-Time Video Streaming using ESP32-CAM” is an IoT-based robotic vehicle designed for remote monitoring and surveillance applications. The robot is capable of streaming live video over Wi-Fi while being controlled wirelessly through a mobile phone or laptop. The system uses the ESP32-CAM module as the main controller and camera unit, along with an L298N motor driver to control DC motors.

The robot creates its own Wi-Fi hotspot, allowing users to connect directly and access a web-based control interface. Through this interface, users can monitor live video and control the movement of the robot in real time. This project demonstrates the integration of Embedded Systems, IoT, Wireless Communication, and Robotics for modern surveillance applications.

Introduction:

With the rapid growth of IoT and robotics technologies, smart surveillance systems are becoming increasingly important in security, military, industrial, and rescue operations. Manual surveillance in dangerous or inaccessible areas may not always be safe or possible. Therefore, robotic surveillance systems provide an efficient and safer alternative.

This project develops a Wi-Fi-controlled surveillance robot using the ESP32-CAM module. The robot can move in different directions and simultaneously transmit live video to a user's device. The ESP32-CAM acts as both a microcontroller and a camera module, reducing overall system complexity and cost.

The project is highly useful in real-time monitoring applications and provides practical knowledge of embedded systems, wireless networking, and robotics.

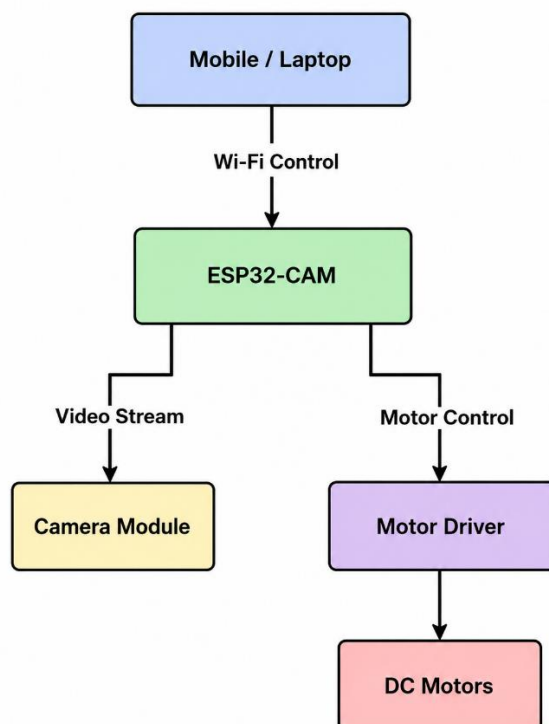
Components Used:

Sr. No.	Component Name	Purpose
1.	ESP32-CAM Module	Main controller and video streaming
2.	L298N Motor Driver	Controls DC motor movement
4.	Wheels	Robot mobility
5.	Battery (7V–12V)	Power supply
6.	Jumper Wires	Electrical connections
7.	Arduino Uno	Used as programmer for ESP32-CAM
8.	Mobile/Laptop	Monitoring and control
9.	Wi-Fi Network	Wireless communication

Objectives:

- To design and develop a wireless surveillance robot
- To implement real-time video streaming using ESP32-CAM
- To remotely control robot movement through Wi-Fi
- To understand IoT and robotics integration
- To create a low-cost and portable surveillance solution
- To provide practical implementation of embedded systems concepts

Block Diagram:



Software Requirements:

- Arduino IDE
- ESP32 Board Package
- AsyncTCP Library
- ESPAsyncWebServer Library
- Embedded C/C++ Programming
- Web Browser for Control Interface

Circuit Connections:

ESP32-CAM Connections

- ESP32-CAM receives power from the motor driver module
- GPIO pins are connected to motor driver input pins
- Camera module is integrated with ESP32 board

Motor Driver Connections

- Motor outputs connected to DC motors
- Input pins connected to ESP32 GPIO pins
- Battery connected to motor driver power input

Programming Connections

Since ESP32-CAM does not have a USB port, an Arduino Uno is used for uploading code:

- UNO TX → ESP32 RX
- UNO RX → ESP32 TX
- GND connected together
- GPIO0 connected to GND during programming mode

Working Principle:

Step 1: Power Supply

The robot is powered using a 7V–12V battery connected to the motor driver module.

Step 2: Wi-Fi Access Point Creation

After power ON, the ESP32-CAM creates its own Wi-Fi hotspot. The user connects their mobile phone or laptop to this Wi-Fi network.

Step 3: Web Server Initialization

The ESP32 runs a web server using the ESPAsyncWebServer and AsyncTCP libraries. A control webpage is hosted internally.

Step 4: Live Video Streaming

The ESP32-CAM captures images continuously and streams live video through the web interface.

Step 5: Robot Control

Control buttons on the webpage send commands such as:

- Forward
- Backward
- Left
- Right
- Stop

The ESP32 processes these commands and controls the motors through the L298N motor driver.

Step 6: Real-Time Surveillance

The robot moves according to user commands while simultaneously transmitting live video feed.

Assembly Procedure:

1. Assemble the 4WD robot chassis
2. Mount DC motors on the chassis
3. Connect wheels to motors
4. Fix the L298N motor driver on the chassis
5. Mount ESP32-CAM securely
6. Connect motors to motor driver outputs
7. Connect ESP32 GPIO pins to driver inputs
8. Provide battery connections
9. Upload program using Arduino Uno
10. Power ON the robot and connect through Wi-Fi

Programming Process:

1. Install Arduino IDE
2. Add ESP32 board package in Board Manager
3. Install required libraries:
 - AsyncTCP
 - ESPAsyncWebServer
4. Select ESP32-CAM board
5. Write/upload Embedded C code
6. Use Arduino Uno as external programmer
7. Open Serial Monitor to obtain IP address
8. Enter IP address in browser for control interface

Features:

- Real-time live video streaming
- Wireless movement control
- Web-based control interface
- Portable robotic design
- ESP32-based compact system
- Wi-Fi hotspot functionality
- Remote surveillance capability

Advantages:

- Low-cost implementation
- Easy wireless control
- Compact and portable
- Useful in dangerous environments
- Reduces human effort
- Real-time monitoring capability
- Easy to assemble and program

Limitations:

- Limited Wi-Fi range
- Video delay due to network conditions
- Battery backup limitations
- Camera quality depends on lighting conditions
- Limited processing power of ESP32

Applications:

- Home security systems
- Industrial surveillance
- Military monitoring
- Rescue operations
- Hazardous area inspection
- Educational robotics projects
- Smart surveillance systems

Important Considerations:

Baud Rate

The baud rate must be set correctly in the code. Incorrect baud rate settings may cause slow or unstable video streaming performance.

Connectivity Issues

If the video feed does not load:

- Disconnect and reconnect to the ESP32 Wi-Fi network
- Refresh the browser page
- Restart the robot if necessary

Safety Precautions

- Never connect 5V directly to 3.3V pins
- Incorrect voltage connections may permanently damage ESP32-CAM
- Ensure proper wiring before powering the system

Future Scope:

- AI-based object detection
- Face recognition system
- Night vision camera support
- Mobile application control
- Cloud video storage
- GPS-based tracking system
- Obstacle avoidance sensors
- Voice-controlled operation

Conclusion:

The “Surveillance Robot with Real-Time Video Streaming using ESP32-CAM” is an

efficient IoT-based robotic monitoring system. The project successfully integrates robotics, embedded systems, wireless communication, and video streaming technologies into a compact and affordable solution.

The robot provides real-time surveillance and remote movement control through Wi-Fi connectivity. It is highly useful in security, industrial, and research applications. This project also provides practical understanding of ESP32 programming, motor control, and IoT-based robotic systems.

References:

1. ESP32 Official Documentation
2. [Arduino Official Website](#)
3. [Random Nerd Tutorials – ESP32-CAM Projects](#)
4. [GeeksforGeeks – IoT and Robotics Concepts](#)
5. [TutorialsPoint – Embedded Systems Tutorials](#)

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